

DR (PROF.) ANIL ARORA PIONEERING ROBOTIC KNEE REPLACEMENT IN INDIA

Dr (Prof.) Anil Arora is the Vice Chairman and Head of Department of Orthopedics & Joint Replacement in Max Superspeciality Hospital, Patparganj, Delhi. He is also the Chief Surgeon & Senior Director of Prof. Arora's Knee & Hip Surgery Clinics, Delhi NCR. As a veteran in the field of joint replacement he has been pioneering the use of technology to provide significant benefit over traditional Joint replacement.

The factors that influence the successful outcome of Total Knee Replacement are accurate alignment, soft tissue preservation and perfect ligament balancing. These are easier said than done. Manual Bony cut and soft tissue balancing may result in poor reproducibility. Incorrect alignment can lead to abnormal early wear, early mechanical loosening of new implanted prosthesis and patello-femoral problems. A functional joint replacement must be well aligned, which implies that it should be along the desired axis and in the correct axial and rotational planes of the joint. There is not much scope for fine tuning of the positioning of the implant. A lot depends on the skill and experience of the operating surgeon.

Robotics is emerging as a very valuable aid in helping surgeons achieve perfection. The robots help create a precise image of the knee that relays patient-specific information to an advanced computer program. The process creates a 3-D model of the knee joint based on CT scan prior to the operation hence making it a customized surgery for each patient. The surgeons use this model

to plan the perfect bone cut, implant positioning, limb alignment, and bone coverage based on the patient's anatomy. The intraoperative robotic device helps to minimize iatrogenic soft tissue and bony injury and improves precision.

Robot supports the surgeon in all these aspects of the surgery and provides near perfect alignment of the joint. Robot basically takes out the guesswork from the surgery and provides better alignment. Prof. Arora says, "Due to



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Professor Arora is blazing a new trail by using Robotic technology to customize joint replacement surgery to suit every patient's unique anatomy, enabling them to enjoy the benefit of a newly replaced joint that feels almost like the original.

the precise nature of the Robotics Assisted Total Knee Replacement, the natural structure of the knee is kept intact, which in turn may lead to a more natural feeling knee and aid in the patient's faster recovery and enjoy a better overall quality of life post-surgery".

Robotic Knee Replacement has been in use in the United States for more than five years. The novel technology is now gaining popularity in India amongst both the Surgeons and patients. Dr Arora who has been evangelizing says, "Robot improves efficiency in the operating room and reduces tissue dissection to perform a knee replacement surgery. Patients treated with this new technology are expected to have better movements and may have a very natural feel of the knee joints".

With the lightning-fast research and developments, robotics has already seen several updates in the last few years. Progressive surgeons like Dr (Prof.) Anil Arora are harnessing it enable people to lead a normal life.

